

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – 26061

Problem Statement Title – AI-Driven Smart Energy Management System for Polar Research Stations

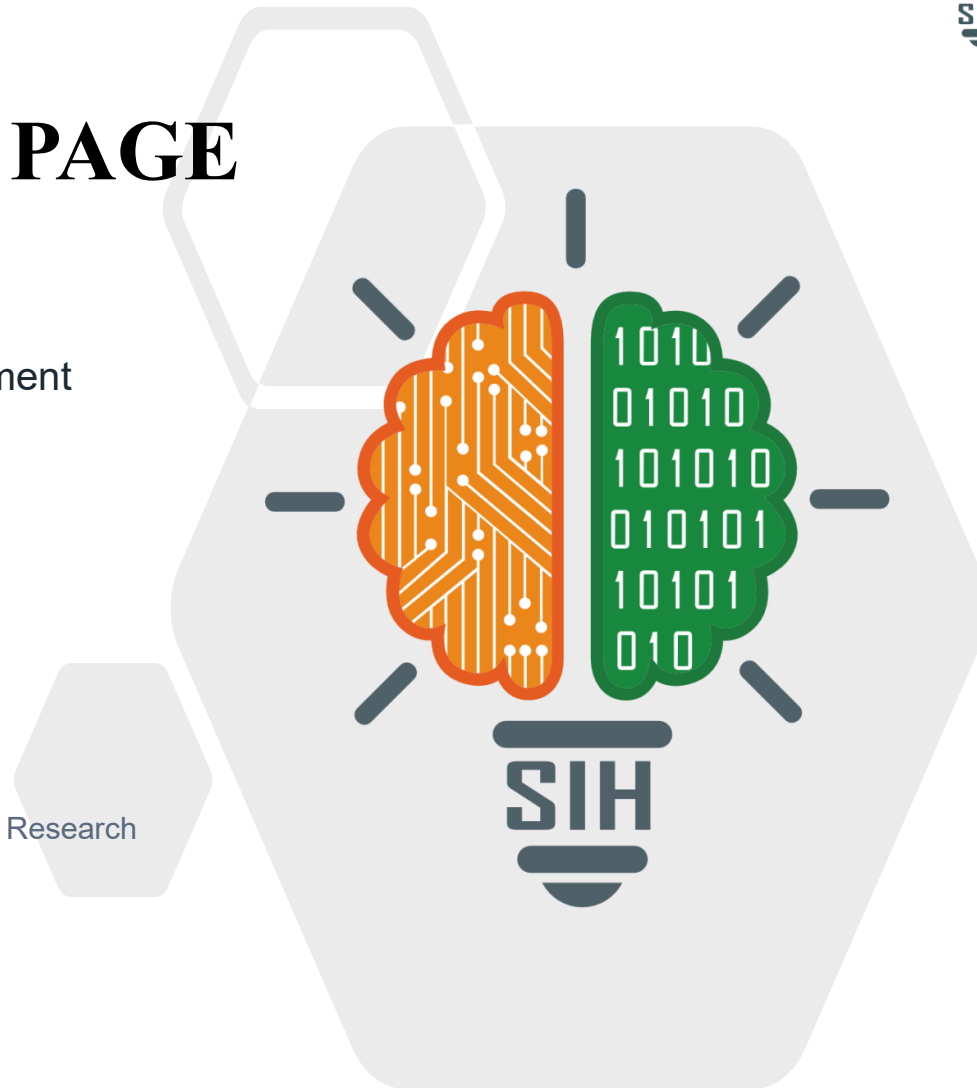
Theme – Clean & Green Technology

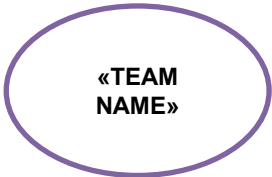
PS Category – Software

Team ID – «TEAM ID»

Team Name – «TEAM NAME»

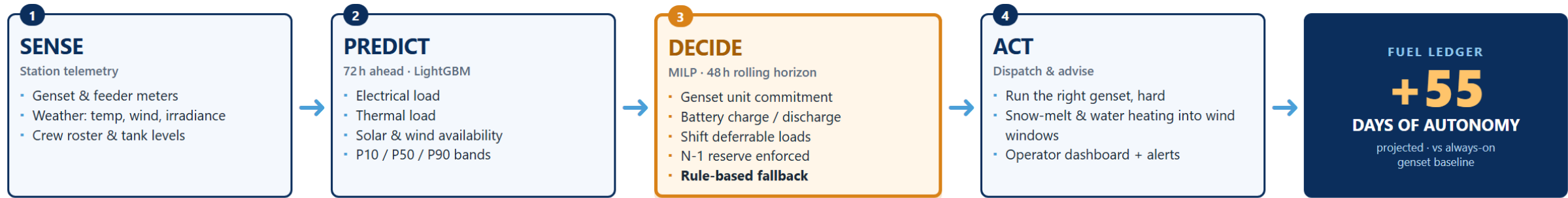
Organisation: Ministry of Earth Sciences · National Centre for Polar and Ocean Research (NCPOR)





IDEA TITLE

HIMSHAKTI | Predictive energy management that turns fuel savings into extra days of survival at Bharati



THE CORE INSIGHT

Genset waste heat warms the station and melts its water — so switching it off to save fuel forces the boiler on. **Electricity and heat are coupled; we co-optimize both.** Electrical-only optimisation gets the wrong answer.

THE PROBLEM

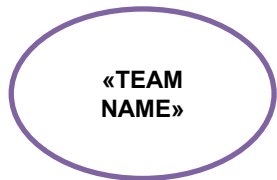
- Bharati, Maitri and Himadri are electrical islands — no grid, no neighbour, no backup supply.
- Diesel arrives ONCE a year by ship. It must last 12 months — run short in winter and nobody is coming.
- A genset at 25 % load burns $\sim 1.5\text{--}2\times$ more fuel per kWh than at 75 %, and wet-stacks below $\sim 30\%$.
- Polar night removes solar for months; blizzards force turbines offline exactly when demand peaks.

OUR SOLUTION

- Forecast electrical load, heat demand, solar and wind 72 h ahead.
- A MILP re-optimises every hour across a 48 h rolling horizon.
- Commits the right genset, holds it in its efficient band, then shuts it off entirely on battery.
- Shifts snow-melting, water heating and charging into wind-surplus windows.
- Tracks the annual fuel budget and reports it as days of autonomy.

WHAT MAKES IT NEW

- Co-optimises electricity AND recovered waste heat. Switching a genset off forces the boiler on — optimising electricity alone gets the wrong answer.
- Predictive horizon optimisation, where today's best polar EMS (Princess Elisabeth Antarctica) uses reactive PLC priority logic.
- Deterministic rule-based fallback — a life-safety system cannot depend on a solver converging.
- Runs offline, CPU-only, on existing station hardware.



TECHNICAL APPROACH



TECHNOLOGIES

- **Core** — Python 3.11
- **Optimisation** — Google OR-Tools CP-SAT (fallback PuLP / CBC)
- **Forecasting** — LightGBM, scikit-learn, pandas, NumPy
- **Backend** — FastAPI + WebSocket
- **Frontend** — React + Vite + Recharts
- **Data** — SQLite / Parquet, no server required
- **Digital twin** — PolarSim, custom physics engine
- **Deployment** — on-premise, CPU-only. No cloud, no GPU, no dependence on satellite uplink
- **Stretch** — ESP32 + CT sensors + relays for a hardware-in-the-loop rig

THE OPTIMISATION CORE

minimise $\Sigma \text{fuel}(g,t) + \text{start-up penalty} + \text{comfort penalty}$

subject to

- **Electrical balance** — $\Sigma P_{\text{gen}} + P_{\text{batt,dis}} + P_{\text{renewable}} = P_{\text{load}} + P_{\text{batt,ch}}$
- **Thermal balance** — $Q_{\text{wasteheat}} + Q_{\text{boiler}} \geq Q_{\text{heating}} + Q_{\text{snowmelt}}$ ← the coupling
- **Genset limits** — minimum stable load (anti wet-stacking), minimum up / down time
- **Storage** — battery SoC bounds with cold-temperature capacity derating
- **Deferrables** — each flexible load must finish inside its deadline window
- **Security** — N-1 spinning reserve, so the largest genset can trip and the station survives

METHODOLOGY — implementation sequence. The optimiser is built and validated on perfect foresight before the forecaster exists, so the hardest component is proven first.

PHASE 1

PolarSim

Physics digital twin of Bharati — a calibrated year at 15 min resolution

PHASE 2

Optimiser + baseline

Rule engine, then MILP. Validated on perfect foresight first

PHASE 3

Forecast models

LightGBM swapped in; measure savings under real forecast error

PHASE 4

Operator dashboard

Live dispatch, fuel burndown, blizzard & failure scenarios

PHASE 5

Field validation

Retrain on real NCPOR telemetry; optional hardware-in-loop rig



FEASIBILITY AND VIABILITY



WHY IT IS FEASIBLE

- Software-only. It retrofits onto the meters and DG control panels a station already has — no new plant, no capital works.
- A MILP of this size solves in seconds on a laptop CPU, so it re-plans every hour without special hardware.
- Fully offline. No reliance on the thin, expensive satellite link.
- Proven precedent: Mawson Station (Australian Antarctic Division) cut diesel from 600,000 L/yr to under 200,000 L/yr through wind integration.
- Aligns with MoES's stated Maitri II objective of a renewable-powered green station.

KEY RISKS → MITIGATION

- ▶ **No public NCPOR energy dataset** → Physics-based digital twin calibrated against published Antarctic station energy data; retrain on real telemetry the moment NCPOR shares it.
- ▶ **Forecast error erodes the saving** → P10/P50/P90 quantile forecasts size a reserve margin. We report savings under real forecast error, never under perfect foresight.
- ▶ **Solver fails to converge** → A deterministic rule-based dispatcher always returns a feasible, safe schedule. The optimiser improves on it; it is never load-bearing for safety.
- ▶ **Asset failure in deep winter** → Mawson lost a turbine in 2017 and never restored it. N-1 reserve is enforced as a hard constraint, not an afterthought.
- ▶ **Operator trust** → Every recommendation is shown with its reason and the litres it saves. The station engineer stays in the loop and can override.

VIABILITY — staged path to deployment. Nothing is automated until it has been shown to work on the station's own data, with the engineer approving every action first.

STEP 1

Validate

Prove the saving on the calibrated digital twin across 10+ weather seeds and a full annual cycle.

STEP 2

Pilot on real logs

Replay NCPOR's historical station records; retrain and re-measure against actual fuel drawn.

STEP 3

Advisory mode

Run live at Bharati recommending only. The station engineer approves or overrides every action.

STEP 4

Closed loop & scale

Automate dispatch once trust is earned; extend to Maitri, Himadri and Maitri II.

IMPACT AND BENEFITS

12 – 18 %

less fuel burned per year

60 – 90 kL

diesel saved annually (1 kL = 1,000 L)

+45 to +65

extra days of autonomy

PROJECTED — basis: ~500,000 L/yr for a Bharati-scale station (Mawson pre-wind: 600,000 L/yr); 12–18 % attributed to predictive dispatch and load shifting. To be replaced by measured simulation results.

OPERATIONAL

Fuel saved converts directly into resupply flexibility. After wind integration, the AAD was able to move Mawson from a mandatory annual refuelling voyage to one every second year. Lower risk of a winter stockout, and schedule slack when sea ice delays a ship.

ENVIRONMENTAL

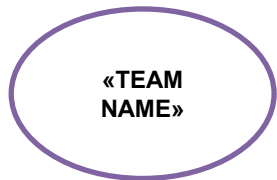
Cuts CO₂ and black carbon. Soot deposition darkens snow, lowers albedo and accelerates local melt — a direct harm in a protected region. Supports India's obligations under the Antarctic Treaty and its Madrid Protocol.

ECONOMIC

A litre of diesel at an Antarctic station costs many times its pump price once ice-class shipping, handling and storage are counted. Mawson's turbines saved AUD 263,000 of fuel in 2014 alone. Fewer voyages compound the saving.

STRATEGIC

A design blueprint for Maitri II's green-station goal, and transferable wherever a microgrid runs on delivered fuel: Himadri, high-altitude border posts, island grids and disaster-relief deployments.



RESEARCH AND REFERENCES



PROBLEM DOMAIN — INDIAN POLAR PROGRAMME

- **NCPOR — Bharati Station**
ncpor.res.in/antarcticas/display/377-bharati
- **NCPOR — Himadri Station**
ncpor.res.in/app/webroot/pages/view/340-himadri-station
- **Smart India Hackathon — PS 26061, MoES / NCPOR**
sih.gov.in

POLICY & ENVIRONMENTAL CONTEXT

- **Antarctic Treaty Secretariat — Protocol on Environmental Protection (Madrid Protocol)**
ats.aq

OPTIMISATION & ML TOOLING

- **Google OR-Tools — CP-SAT solver**
developers.google.com/optimization/cp
- **LightGBM — gradient boosting framework**
lightgbm.readthedocs.io

PRECEDENT — POLAR STATION ENERGY SYSTEMS

- **Australian Antarctic Program — Wind power at Antarctic stations**
antarctica.gov.au/antarctic-operations/stations-and-field-locations/amenities-and-operations/renewable-energy/wind-power/
- **Mawson: Antarctica's first wind-powered station — AAD magazine**
antarctica.gov.au/magazine/issue-6-autumn-2004/feature/mawson-antarcticas-first-wind-powered-station/
- **Princess Elisabeth Antarctica — renewable energy and smart microgrid (Laborelec / Schneider Electric EMS)**
antarcticstation.org/station/renewable_energies
- **RMI — Renewables at the South Pole**
rmi.org/blog_2015_11_11_renewables_at_the_south_pole/

KEY FINDING APPLIED IN THIS PROPOSAL

Mawson's diesel use fell from 600,000 L/yr to under 200,000 L/yr, letting the AAD refuel every second year instead of annually. Fuel efficiency converts directly into resupply resilience — which is exactly the outcome we optimise for.